TRIPATHI STUDY ZONE

Jee Main 2020(Sep)

02-Sep-2020 (Morning Shift)

Question Paper, Key and Solutions

TRIPATHI STUDY ZONE
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TRIPATHI STUDY ZONE
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Physics

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. An amplitude modulated wave is represented by the expression $v_m = 5(1+0.6\cos 6280t)\sin\left(211\times10^4 t\right)$ volts. The minimum and maximum amplitudes of the amplitude modulated wave are, respectively

1) $\frac{5}{2}V, 8V$ 2) 3V, 5V 3) 5V, 8V 4) $\frac{3}{2}V, 5V$

Key: 1

Sol: Minimum amplitude $5(1-0.6)=2$

Maximum amplitude $5(1+0.6)=8$

No correct options

2. Two identical strings X and Z made of same material have tension T_X and T_Z in them. If their fundamental frequencies are 450Hz and 300Hz respectively, then the ratio T_X/T_Z is

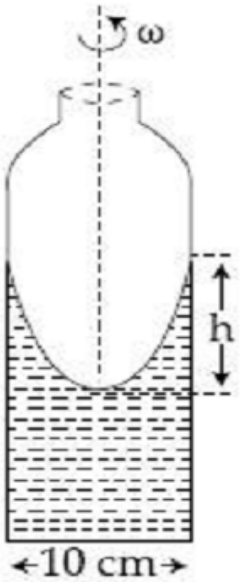
1) 2.25 2) 1.5 3) 0.44 4) 1.25

Key: 1

Sol: $f = \frac{V}{2l} = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$

For identical strings $\frac{f_x}{f_z} = \sqrt{\frac{T_X}{T_Z}} \Rightarrow \frac{T_X}{T_Z} = 2.25$

3. A cylindrical vessel containing a liquid is rotated about its axis so that the liquid rises at its sides as shown in the figure. The radius of vessel is 5cm and the angular speed of rotation is ω rad s^{-1}. The difference in the height, h(in cm) of liquid at the centre of vessel and at the side will be:



1) $\frac{25\omega^2}{2g}$ 2) $\frac{2\omega^2}{5g}$ 3) $\frac{5\omega^2}{2g}$ 4) $\frac{2\omega^2}{25g}$

Key: 1

Sol: $sgh = \int_h^r s\omega^2 x dx = \frac{s\omega^2 r^2}{2}$

$\Rightarrow h = \frac{\omega^2 r^2}{2g} = \frac{25r^2}{2g}(in\,CGS)$

4. Magnetic materials used for making permanent magnets (P) and magnets in a transformer (T) have different properties of the following, which property best matches for the type of magnet required?

1) P: Small retentivity, large coercivity

2) T: Large retentivity, large coercivity

3) P: Large retentivity, large coercivity

4) T: Large retentivity, small coercivity

Key: 3

Sol: for permanent magnet retentivity must be high. For magnets in transformer hysteresis loss must be small

5. A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Assuming the gases to be ideal and the oxygen bond to be rigid, the total internal energy (in units of RT) of the mixture is

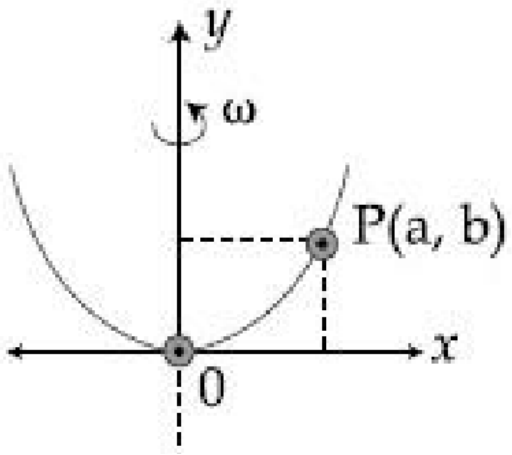
1) 11 2) 15 3) 13 4) 20

Key: 2

Sol: oxygen bonds are rigid $\Rightarrow$ no vibrational degrees of freedom.

$U = 3\left(\frac{5}{2}RT\right) + 5\left(\frac{3}{2}RT\right) = 15RT$

6. A bead of mass m stays at points P(a,b) on a wire bent in the shape of a parabola $y = 4Cx^2$ and rotating with angular speed ω (see figure). The value of ω is (neglect friction):



1) $2\sqrt{gC}$ 2) $\sqrt{\frac{2g}{C}}$ 3) $2\sqrt{2gC}$ 4) $\sqrt{\frac{2gC}{ab}}$

Key: 3

Sol: $y = 4cx^2$

$\frac{dy}{dx} = 8cx = \tan\theta = \frac{\omega^2 r}{g}(r = x)$

$\Rightarrow \omega^2 = 8c.g$

7. Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity ρ_C, ρ_T, ρ_M and ρ_A respectively.

Then

1) $\rho_C > \rho_A > \rho_T$ 2) $\rho_A > \rho_M > \rho_C$ 3) $\rho_M > \rho_A > \rho_C$ 4) $\rho_A > \rho_T > \rho_C$

Key: 3

Sol: data memory based question

8. In a reactor, 2kg of $_{92}U^{235}$ fuel is fully used up in 30 days. The energy released per fission is 200 MeV. Given that the Avogadro number, $N = 6.023 \times 10^{26}$ per kilo mole and 1 $eV = 1.6 \times 10^{-19} J$. The power output of the reactor is close to

1) 35MW 2) 60MW 3) 54MW 4) 125MW

Key: 2

Sol: $\text{Total energy} = \dfrac{\text{mass}}{\text{molar mass}} \times \text{avogadro no} \times 200\text{e} \times 10^6 \text{J}$

$$\text{power} = \frac{\text{total energy}}{30\,\text{days}} = \frac{2 \times 6.023 \times 10^{23} \times 200 \times 1.6 \times 10^{-13}}{235 \times 10^{-3} \times 30 \times 24 \times 3600} \text{W}$$

$$= 6.33 \times 10^7 \text{W}$$

9. Train A and train B are running on parallel tracks in the opposite directions with speeds of 36km/hour and 72km/hour respectively. A person is walking in train A in the direction opposite to its motion with a speed of 1.8km/hour. Speed (in ms^{-1}) of this person as observed from train B will be close to: (take the distance between the tracks as negligible)

1) $29.5 ms^{-1}$ 2) $28.5 ms^{-1}$ 3) $31.5 ms^{-1}$ 4) $30.5 ms^{-1}$

Key: 1

Sol: $V_A = 10 m/s$ $V_B = -20 m/s$

Velocity of person w.r.t train A, $V_{PA} = -0.5 m/s$

$\therefore$ Velocity of person

$V_P = V_{PA} + V_A = 9.5 ms$

And $V_{PB} = V_P - V_B = 9.5 - (-20) = 29.5 m/s$

10. The mass density of a spherical galaxy varies as $\frac{K}{r}$ over a large distance 'r' from its centre. In that region, a small star is in a circular orbit of radius R. Then the period of revolution, T depends on R as

1) $T \propto R$ 2) $T^2 \propto \frac{1}{R^3}$ 3) $T^2 \propto R^3$ 4) $T^2 \propto R$

Key: 4

Sol: Gravitational force on star of mission

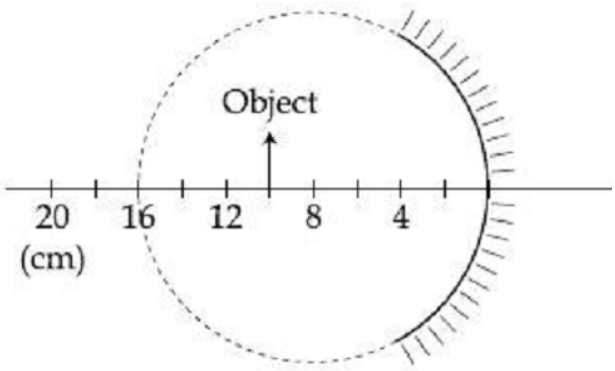
$F_G = \frac{Gmn}{r^2}$ = where M is mass of galaxy

$So, M = \int_0^r 4\pi r^2 dr \frac{K}{r} = \frac{4\pi k r^2}{2}$

$F_a = m\omega^2 r = m.r\frac{4\pi^2}{T^2}$

$\Rightarrow T^2 = \frac{4\pi^2 mr}{F_4} = \frac{4\pi^2 mr}{Gm/r^2} = \frac{4\pi^2 r^3}{G\left(\frac{4\pi k r^3}{2}\right)} \Rightarrow T \propto \sqrt{r}$

11. A spherical mirror is obtained as shown in the figure from a hollow glass sphere. If an object is positioned in front of the mirror, what will be the nature and magnification of the image of the object? (Figure drawn as schematic and not to scale)



1) Erect, virtual and unmagnified 2) Erect, virtual and magnified

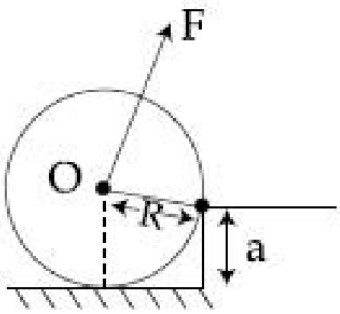
3) Inverted, real and magnified 4) Inverted, real and unmagnified

Key: 4

Correct Ans: 4

Sol: Object is beyond centre of curvature hence image is real, inverted and diminished

12. A uniform cylinder of mass M and radius R is to be pulled over a step of height a (a<R) by applying a force F at its centre 'O' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is:



1) $Mg\sqrt{1-\frac{a^2}{R^2}}$

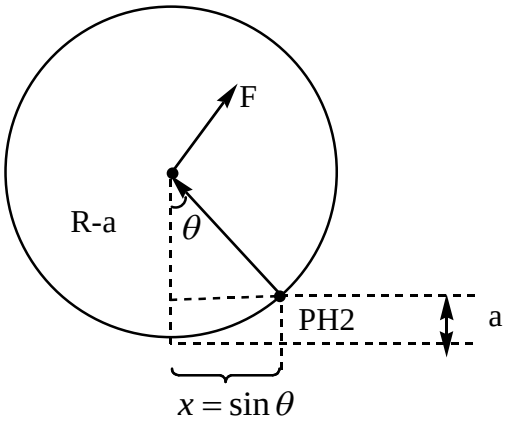
2) $Mg\sqrt{1-\left(\frac{R-a}{R}\right)^2}$

3) $Mg\sqrt{\left(\frac{R}{R-a}\right)^2-1}$

4) $Mg\frac{a}{R}$

Key: 2

Sol: Assume no slipping about P

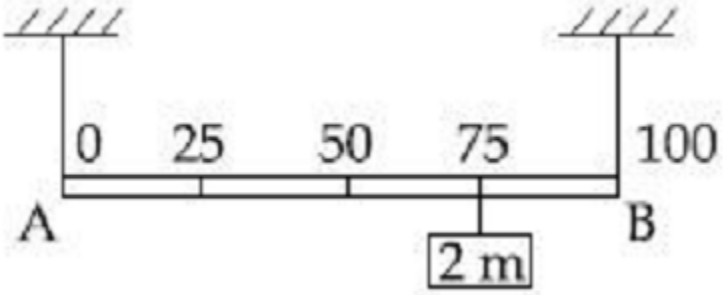


Torque of F about P> torque of gravity

$\Rightarrow FR > Mgx = MgR\sin\theta$

$\Rightarrow F_{min} = Mg\sin\theta = Mg\sqrt{1-\left(\frac{R-a}{R}\right)^2}$

13. Shown in the figure is rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass 'm' and has another weight of mass 2m hung at a distance of 75cm from A. The tension in the string at A is



1) 2mg 2) 1mg 3) 0.5 mg 4) 0.75mg

Key: 2

Sol: Taking torque about point B =0

$T_A = 100 = mg.50 + 2mg.25$

$\Rightarrow T_A = mg$

14. If speed V, area A and force F are chosen as fundamental units, then the dimension of Young's modulus will be:

1) $FA^{-1}V^0$ 2) FA^2V^{-2} 3) FA^2V^{-3} 4) FA^2V^{-1}

Key: 1

Sol: Using formulae for longitudinal stress,

$\frac{F}{A} = Y\frac{\Delta l}{l}$

$\Rightarrow [Y] = [F].[A]^{-1}$

15. Interference fringes are observed on a screen by illuminating two thin slits 1mm apart with a light source $(\lambda = 632.8\,nm)$. The distance between the screen and the slits is 100cm. If a bright fringe is observed on a screen at a distance of 1.27mm from the central bright fringe, then the path difference between the waves, which are reaching this point from the slits is close to:

1) 2nm 2) 1.27μm 3) 2.87μm 4) 2.05μm

Key: 2

Sol: Path difference $= d\sin\theta = d.\frac{y}{D}$

$= \frac{(1mm)(1.27)mm}{1mm} = 1.27\mu m$

16. A plane electromagnetic wave, has frequency of 2.0×10^{10} Hz and its energy density is $1.02\times10^{-8}\,J/m^3$ in vacuum. The amplitude of the magnetic field of the wave is close to $\left(\frac{1}{4\pi\varepsilon_0}=9\times10^9\frac{Nm^2}{C^2}\text{and speed of light}=3\times10^8\,ms^{-1}\right)$:

1) 190nT 2) 180nT 3) 160nT 4) 150nT

Key: 3

Sol: Energy density $=\frac{B_0^2}{2\mu_0}=\frac{B_0^2\,\epsilon_0}{2\mu_0\,\epsilon_0}=\frac{B_0^2C^2\,\epsilon_0}{2}$

$as\left(C^2=\frac{1}{\mu_0\,\epsilon_0}\right)\quad\Rightarrow B_0=\frac{1}{C}\sqrt{\frac{2(\text{Energy density})}{\epsilon_0}}$

$=\frac{1}{3\times10^8}\sqrt{\frac{2\times1.02\times10^{-8}}{4\pi\,\epsilon_0}4\pi}$

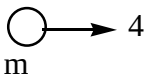
$=\frac{1}{3\times10^8}\sqrt{2\times.02\times10^{-8}\times9\times10^9\times4\pi}\quad=158.5\times10^{-9}T$

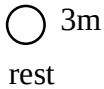
17. A particle of mass m with an initial velocity $u\hat{i}$ collides perfectly elastically with a mass 3m at rest. It moves with a velocity $v\hat{j}$ after collision, then, v is given by:

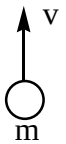
1) $v=\sqrt{\frac{2}{3}}u$ 2) $v=\frac{1}{\sqrt{6}}u$ 3) $v=\frac{u}{\sqrt{3}}$ 4) $v=\frac{u}{\sqrt{2}}$

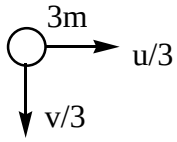
Key: 4

Sol:

before collision 



after collision 

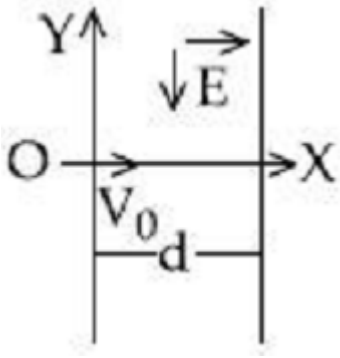


where we have used moment conservative

using energy concentrate

$\frac{1}{2}mv^2=\frac{1}{2}3m\left\{\left(\frac{v}{3}\right)^2+\left(\frac{4}{3}\right)^2\right\}\quad\Rightarrow V^2=\frac{V^2}{3}+\frac{4^2}{3}\Rightarrow\frac{2V^2}{3}=\frac{4^2}{3}\quad\Rightarrow V=4/\sqrt{2}$

18. A charged particle (mass m and charge q) moves along X axis with velocity V_0. When it passes through the origin it enters a region having uniform electric field. $\vec{E} = -E\hat{j}$ which extends upto $x = d$. Equation of path of electron in the region $x > d$ is



1) $y = \frac{qEd^2}{mV_0^2}x$

2) $y = \frac{qEd}{mV_0^2}x$

3) $y = \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$

4) $y = \frac{qEd}{mV_0^2}(x - d)$

Key: 3

Sol: $V_x = V_0$

$V_y = \left(\frac{qE}{m}\right).\frac{d}{V_0}$

when $x = d, y = -\frac{1}{2}\left(\frac{qe}{m}\right)\left(\frac{d}{v_0}\right)^2 = -y_0(sum)$

Now for $(x > n_0)$

$y - y_0 = -V_y t$

$x - d = V_x t = V_0 t$

eliminating t,-$\left(\frac{y - y_0}{V_y}\right) = \frac{x - d}{V_0}$

$\Rightarrow y = \frac{d - x}{V_0}V_y + y_0 = \frac{d - x}{V_0^2}.\frac{qEd}{m} + \frac{qEd^2}{2mV_0^2}$ $= \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$

19. A beam of protons with speed $4 \times 10^5 ms^{-1}$ enters a uniform magnetic field of 0.3T at an angle of 60º to the magnetic field. The pitch of the resulting helical path of protons is close to: (Mass of the proton 1.67×10^{-27} kg, charge of the proton $= 1.69 \times 10^{-19}$ C)

1) 12cm 2) 4cm 3) 2cm 4) 5cm

Key: 2

Sol: $pitch = V\cos\theta.T = V\cos\theta \times \frac{2\pi m}{qB}$

$$\frac{4\times10^{5}\times\cos60^{o}\times2\pi\times1.67\times10^{-27}}{1.69\times10^{-19}\times0.3}$$

$$=\frac{1.67\times4\pi}{1.69\times0.3}\times10^{-3}m\simeq4cm$$

20. The least count of the main scale of a vernier callipers is 1mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1cm and 3.2cm and 4th VSD coincides with a main scale division. The length of the cylinder is (VSD is vernier scale division)

1) 3.07cm 2) 3.2cm 3) 2.99cm 4) 3.21cm

Key: 1

Sol: least count = $msD - vsD = 0.1mm$

Positive zero error $= 7\times(lc) = 0.7mm$

Measured length $31mm + 4(V.C) = 31.4mm$

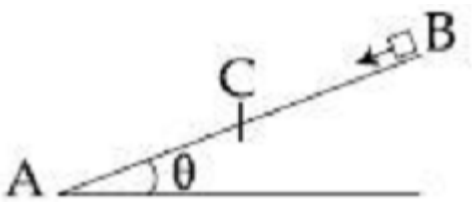
$l = 31.4mm - 0.7mm = 30.7mm$

(NUMERICAL VALUE TYPE)

This section contains 5 questions. Each question is numerical value. For each question, enter the correct numerical value(in decimal notation, truncated/rounded-off to second decimal place.(e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. A small block starts slipping down from a point B on an inclined plane AB, which is making an angle θ with the horizontal section BC is smooth and the remaining section CA is rough with a coefficient of friction μ. It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If BC=2AC, the coefficient of friction is given by $\mu = k\tan\theta$. The value of k is_____



Key: 3.00

Sol: Work done by gravity in moving from B to A = mg sing θ (BA)
Work done by friction in moving from C to A $= \mu mg\cos\theta(CA)$
Total work $= \Delta KE = 0$
$\Rightarrow mg\sin\theta.(BA) - \mu mg\cos\theta CA = 0$
$\Rightarrow \mu = \tan\theta\left(\frac{BA}{CA}\right) = 3\tan\theta$

22. A $5\mu F$ capacitor is charged fully by a 220V supply. It is then disconnected from the supply and is connected in series to another uncharged $2.5\mu F$ capacitor. If the energy change during the charge redistribution is $\frac{X}{100}$J then value of X to the nearest integer is_____.

Key: 36.00

Correct Ans: 4.00

Sol: lost during charge redistribution
$= \frac{1}{2}\frac{C_1C_2}{C_1+C_2}V^2 = \frac{1}{2}\frac{5\times 2.5}{5+2.5}\times(220)^2\times 10^{-6}$
$= \frac{5}{6}\times(2.2)^2\times 10^{-2} = \frac{X}{100}$
$\Rightarrow x = \frac{5\times(2.2)^2}{6} = 4.03$

23. When radiation of wavelength λ is used to illuminate a metallic surface, the stopping potential is V. When the same surface is illuminated with radiation of wavelength 3λ, the stopping potential is $\frac{V}{4}$. If the threshold wavelength for the metallic surface is $n\lambda$ then value of n will be

Key: 9.00

Sol: $\left.\begin{aligned} eV &= \frac{hC}{\lambda} - \phi \\ \frac{eV}{4} &= \frac{hC}{3\lambda} - \phi \end{aligned}\right\}$ solving $\frac{hC}{\lambda} - \phi = 4\left(\frac{hC}{3\lambda} - \phi\right)$

$\phi = \frac{hC}{h\lambda} \Rightarrow n = 9 \qquad \phi = \frac{hC}{9\lambda}$

24. An engine takes in 5 moles of air at 20°C and 1atm, and compresses it adiabatically to $1/10^{th}$ of the original volume. Assuming air to be a diatomic ideal gas made up of rigid molecules, the change in its internal energy during this process comes out to be X kJ. The value of X to the nearest integer is_____

Key: 46.00

Sol: $PV^{Y} = cont, Y = \frac{7}{5}$

$\Rightarrow TV^{2/5} = cont$

$T_{fin}\left(\frac{V}{10}\right)^{2/5} = T_m V^{2/5} \Rightarrow T_{fin} = 293\left(10^{2/5}\right)$

$\Delta U = \frac{nR\Delta T}{\gamma - 1} = \frac{5R}{2/5}.293\left(10^{2/5} - 1\right) = \frac{25R}{2}\left(10^{2/5} - 1\right) \times 293 = 30.5kJ\left(10^{2/5} - 1\right)$

25. A circular coil of radius 10cm is placed in a uniform magnetic field of $3.0 \times 10^{-5} T$ with its plane perpendicular to the field initially. It is rotated at constant angular speed about an axis along the diameter of coil and perpendicular to magnetic field so that it undergoes half of rotation in 0.2s. The maximum value of EMF induced $(\text{in } \mu\text{V})$ in the coil will be close to the integer

Key: 15.00

Sol: $\phi = B.\pi r^2 \cos\omega t$

$\varepsilon = B\omega\pi r^2 \sin\omega t$

$\varepsilon_0 = B\omega\pi r^2$

$= 3 \times 10^{-5} \times \left(\frac{\pi}{0.2}\right) \times \pi \times 10^{-2}$

$= \frac{3\pi^2}{2}\mu V \simeq 15\mu V$

CHEMISTRY

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Which of the following compounds will show retention in configuration on nucleophilic substitution by OH^- ion?

1) $CH_3-\underset{\underset{C_2H_5}{|}}{CH}-CH_2Br$

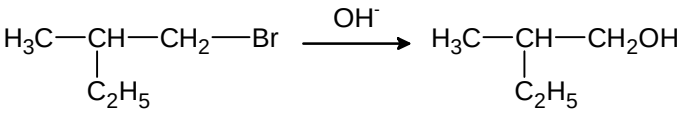
2) $CH_3-\underset{\underset{CH_3}{|}}{CH}-Br$

3) $CH_3-\underset{\underset{C_6H_{13}}{|}}{\overset{\overset{Br}{|}}{C}}-H$

4) $CH_3-\underset{\underset{C_6H_5}{|}}{CH}-Br$

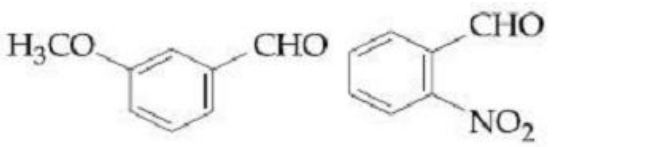
Key: 1

Sol: If the chiral centre is not involved in the reaction option 1 is the answer.



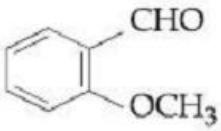
Configuration is retained

2. The increasing order of the following compounds towards HCN addition is:

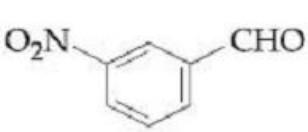


(i)

(ii)



(iii)



(iv)

1) iii<iv<ii<i 2) i<iii<iv<ii 3) iii<i<iv<ii 4) iii<iv<i<ii

Key: 3

Sol: $-NO_2$ being a –R group increases the reactivity. When it is at ortho position. At meta position it shows –I effect which is more than the –I effect of –OMe. At ortho position –OMe shows +R effect, then decreasing the reactivity.
So the order is (ii) > (iv) > (i)> (iii).

3. If AB_4 molecules is a polar molecule, a possible geometry of AB_4 is:

1) Square planar 2) Rectangular planar

3) Tetrahedral 4) Square pyramidal

Key: 4

Sol: When AB_4 molecule is present in square pyramidal structure will have dipole moment. The other three structures do not have dipole moment

4. For the following Assertion and Reason, the correct option is

Assertion (A): When Cu(II) and sulphide ions are mixed, they react together extremely quickly to give a solid

Reason (R): The equilibrium constant of $Cu^{2+}(aq) + S^{2-}(aq) \longrightarrow CuS(s)$ is high because the solubility product is low.

1) Both A and R are true but R is not the explanation for A

2) Both A and R are true and R is the explanation for A

3) A is false and R is true.

4) Both A and R are false.

Key: 1

Sol: Precipitation reactions involving ions are very fast. The equilibrium constant for the formation of CuS is high as it is precipitated in acidic medium. Both A and R are correct. But higher equilibrium constant doesn't mean higher rate.

5. The metal mainly used in devising photoelectric cells is

1) Na 2) Cs 3) Rb 4) Li

Key: 2

Sol: Due to its low ionization energy ceasium is used in photo electric cells

6. Consider the following reactions:

(i) Glucose+ROH $\xrightarrow{dry\ HCl}$ *Acetal* $\xrightarrow[(CH_3CO)_2O]{x.eq.of}$ *acetyl derivative*

(ii) Glucose $\xrightarrow{Ni/H_2}$ *A* $\xrightarrow[(CH_3CO)_2O]{y\ eq.of}$ *acetyl* derivative

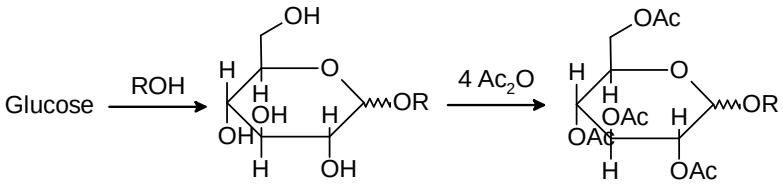
(iii) Glucose $\xrightarrow[(CH_3CO)_2O]{z\ eq.of}$ *acetyl* derivative

'*x*','*y*' and 'z' in these reactions are respectively.

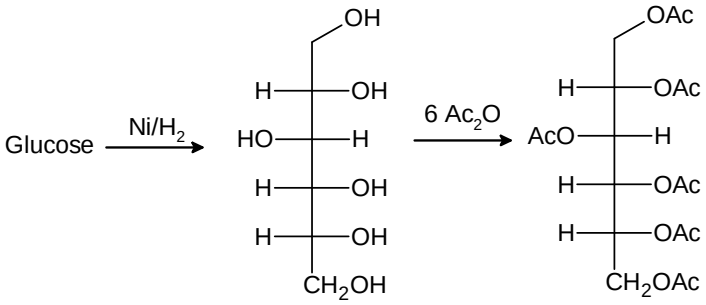
1) 4,6 and 5 2) 5,6 and 5 3) 5,4 and 5 4) 4,5 and 5

Key: 1

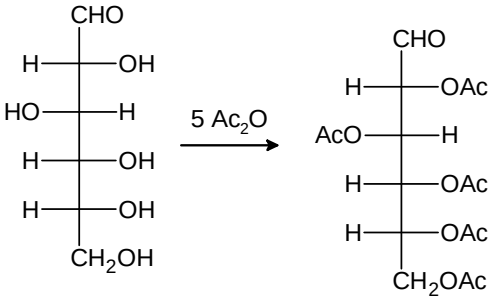
Sol:



$\Rightarrow x = 4$



$\Rightarrow y = 6$



7. While titrating dilute HCl solution with aqueous NaOH, which of the following will not be required?

1) Pipette and distilled water

2) Clamp and phenolphthalein

3) Bunsen burner and measuring cylinder

4) Burette and porcelain tile

Key: 3

Sol: During the titration of HCl Vs NaOH except Bunsen burner and measuring cylinder all the other given are used

8. Consider that a d^6 metal ion $\left(M^{2+}\right)$ forms a complex with aqua ligands, and the spin only magnetic moment of the complex is 4.90BM. The geometry and the crystal field stabilization energy of the complex is

1) tetrahedral and $-1.6\Delta_t + 1P$

2) octahedral and $-24\Delta_0 + 2P$

3) tetrahedral and $-0.6\Delta_t$

4) octahedral and $-1.6\Delta_0$

Key: 3

Sol: The metal ion M^{2+} with d^6 configuration with aqua ligands will have $t_{2g}^4 eg^2$ in octahedral field and $e^3 t_2^3$. Then crystal field stabilization energy in octahedral field will be $\left[\left(-0.4\times 4\right)+\left(0.6\times 2\right)+P\right]\Delta_0$ and in tetrahedral field $\left[\left(-0.6\times 3\right)+0.4\times 3+P\right]\Delta_t$. The tetrahedral value is $-0.6\Delta_t + p$; the value match with 3

9. On heating compound (A) gives a gas (B) which is a constituent of air. This gas when treated with H_2 in the presence of a catalyst gives another gas (C) which is basic in nature. (A) should not be:

1) $\left(NH_4\right)_2 Cr_2O_7$ 2) NaN_3 **3)** $Pb\left(NO_3\right)_2$ 4) NH_4NO_2

Key: 3

Sol: The gas present in air that can react with hydrogen in the presence of catalyst to form a basic gas $\left(NH_3\right)$ is N_2. The N_2 will be given by heating all the three compounds except $Pb\left(NO_3\right)_2$

10. The statements that is not true about ozone is:

1) it is a toxic gas and its reaction with NO gives NO_2.

2) in the stratosphere, it forms a protective shield against UV radiation.

3) In the atmosphere, it is depleted by CFCs.

4) In the stratosphere, CFCs release chlorine free radicals (Cl) which reacts with O_3 to give chlorine dioxide radicals.

Key: 4

Sol: Chlorine react with ozone forming $ClO^{\bullet}$ Free radical but not $ClO_2^{\bullet}$ free radical

$O_3 + Cl^{\bullet} \rightarrow ClO^{\bullet} + O_2$

The other statements are correct

11. For octahedral Mn(II) and tetrahedral Ni(II) complexes, consider the following statements:

I) both the complexes can be high spin.

II) Ni(II) complex can very rarely below spin.

III) with strong field ligands, Mn(II) complexes can below spin.

IV) aqueous solution of Mn(II) ions is yellow in colour.

The correct statements are:

1) II,III and IV only

2) I and II only

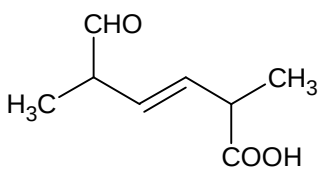
3) I, II and III only

4) I, III, and IV only

Key: 3

Sol: Mn^{2+} with d^5 configuration form paramagnetic in the presence of both weak and strong field. If can form both high spin and low spin complexes. Similary Ni^{2+} with d^8 configuration can form both low spin and high spin complexes. Except with very strong ligands such as CN^-, Co, Ni^{2+} always form tetrahedral complexes with high spin. Aqueous solutions of $Mn(II)$ almost colourless.

12. The IUPAC name for the following compounds



1) 6- formyl-2-methyl-hex-3-enoic acid

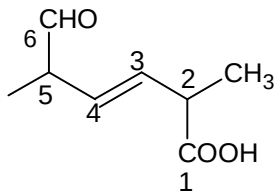
2) 2,5-dimethyl-6-carboxy-hex-3-enal

3) 2,5- dimethyl – 6 – oxo- hex-3-enoic acid

4) 2,5- dimethyl -5- carboxy-hex-3- enal

Key: 3

Sol:



2, 5-dimethyl-6-oxo Lex-3-enoic acid

13. An open beaker of water in equilibrium with water vapour is in a sealed container. When a few grams of glucose are added to the beaker of water, the rate at which water molecules.

1) leaves the solution decreases

2) leaves the vapour decreases

3) leaves the vapour increases

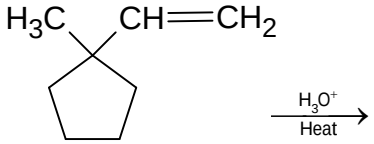
4) leaves the solution increases

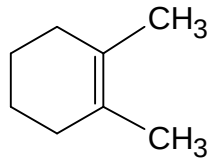
Key: 3

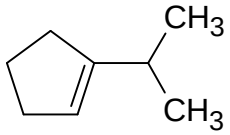
Correct Key: 1

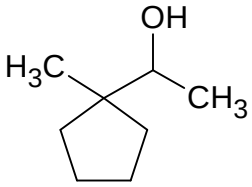
Sol: Addition of glucose to water decreases the concentration of water molecules in the surface which leads to lowering of rate of evaporation. ie, the rate at which water molecule leaves the solution decreases.

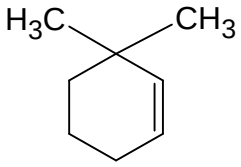
14. The major product following the reaction is



1) 

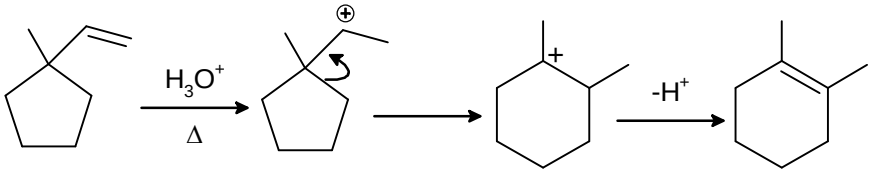
2) 

3) 

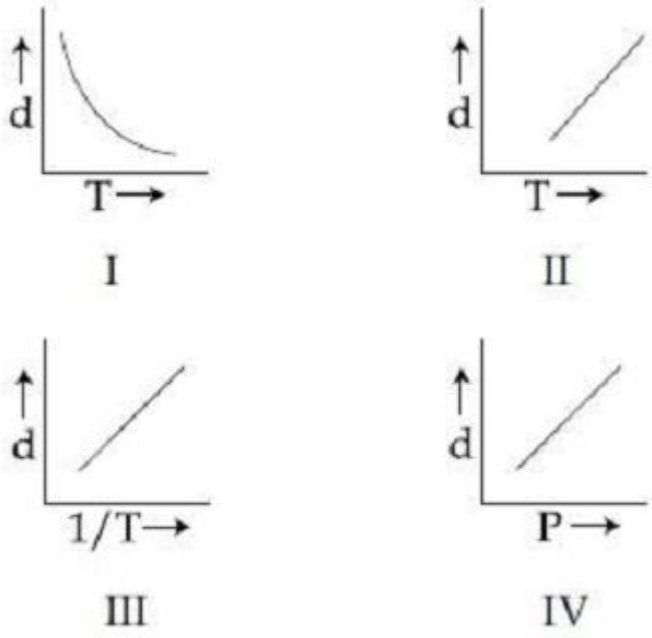
4) 

Key: 1

Sol:



15. Which one of the following graphs is not correct for ideal gas?



D=Density, P=Pressure, T=Temperature

1) I 2) IV 3) II 4) III

Key: 3

Sol:

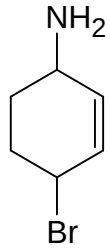
$PM = dRT$

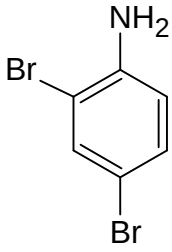
$\Rightarrow d \propto \frac{1}{T}$

Also $\Rightarrow d \propto P$

$\Rightarrow$ I, III & IV are correct

16. In Carius method of estimation of halogen, 0.172g of an organic compound showed presence of 0.08g of bromine. Which of these is the correct structure of the compound?

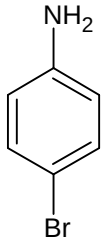
1) 

2) $H_3C - CH_2 - Br$ **3)** 

4) $H_3C - Br$

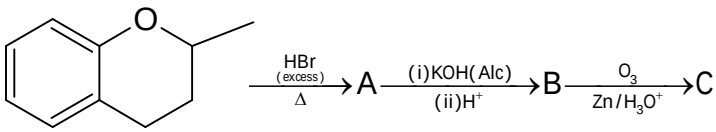
Key: 1

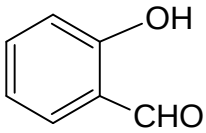
Sol: $\%\ Br = \frac{0.08}{0.172} \times 100 = 46.51$

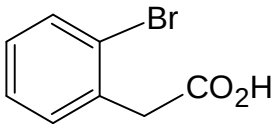
For 

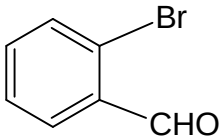
, $\%Br = \frac{80}{80+72+14+6} \times 100 = 46.51$

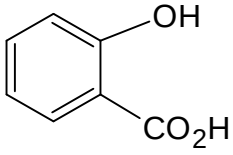
17. The major aromatic product C in the following reaction sequence will be



1) 

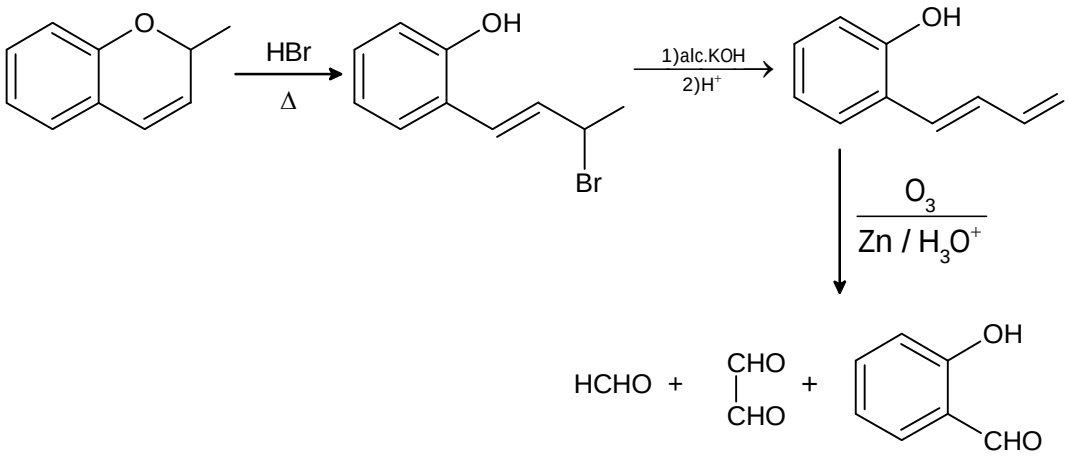
2) 

3) 

4) 

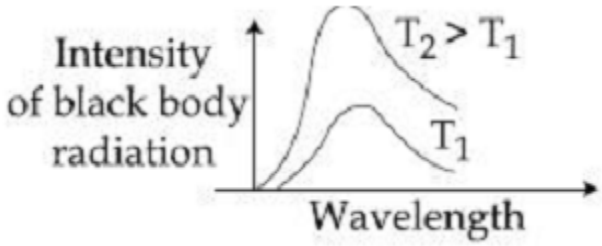
Key: 1

Sol:

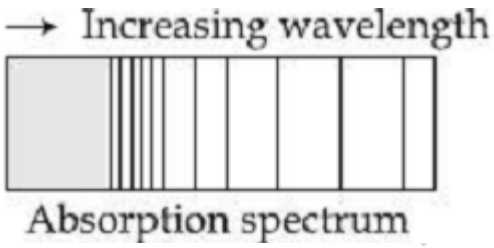


18 The figure that is not a direct manifestation of the quantum nature of atoms is

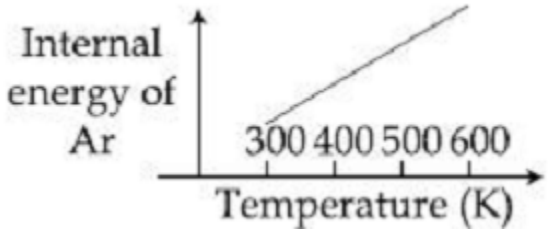
1)



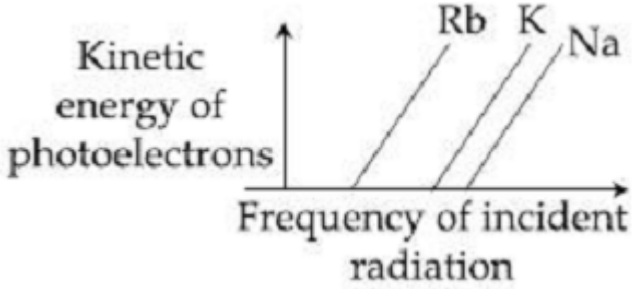
2)



3)



4)



Key: 3

Sol: * Intensity of black body reaction is explained by Plank's Quantum Theory.
* Absorption spectrum of photo electric effect are also explained by quantum theory.

19. In general, the property (magnitudes only) that shows an opposite trend in comparison to other properties across a period is

1) Atomic radius 2) Electron gain enthalpy

3) Electronegativity 4) Ionization enthalpy

Key: 1

Sol: In a period from left to right generally atomic radius decreases but other properties such as Electron gain enthalpy, Electronegativity, Ionization enthalpy increases.

20. Which of the following is used for preparation of colloids?

1) Ostwald Process 2) Bredig's Arc Method

3) Van Arkel Method 4) Mond Process

Key: 2

Sol: Bredig's Arc method is used to prepare colloids of noble metal.

(NUMERICAL VALUE TYPE)

This section contains 5 questions. Each question is numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.(e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).
Marking scheme: +4 for correct answer , 0 if not attempted and 0 in all other cases.

21. The oxidation states of iron atoms in compound (A), (B) and (C), respectively, are x,y and z. The sum of x,y and z is_____

$Na_4\left[Fe(CN)_5(NOS)\right]$ (A) $Na_4\left[FeO_4\right]$ (B) $\left[Fe_2(CO)_9\right]$ (C)

Key: 6

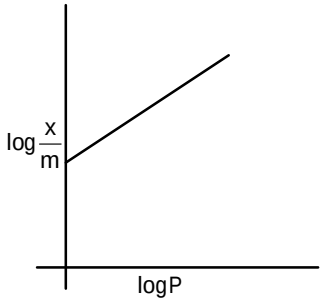
Sol: Oxidation state Fe
in $Na_4\left[Fe(CN)_5(NO_5)\right]$ $is+2$
in Na_4FeO_4 is $+4$
in $Fe_2(CO)_9$ is 0
So total 6

22. The mass of gas adsorbed, x, per unit mass of adsorbate, m, was measured at various pressures, p. A graph between $\log\frac{x}{m}$ and $\log p$ gives a straight line with slope equal to 2 and the intercept equal to 0.4771. The value of $\frac{x}{m}$ at a pressured of 4 atm is : (Given $\log 3 = 0.4771$)

Key : 6.00

Correct Key : 48.00

Sol:



$$\frac{x}{m} = k.p^{1/n}$$

$$\log\frac{x}{m} = \log k + \frac{1}{n}\log p$$

$$\frac{1}{n} = 2$$

$$\log k = 0.4771$$

$$k = 3$$

$$\Rightarrow \frac{x}{m} = 3\times 4^2 = 3\times 16 = 48$$

23. The internal energy change (in J) when 90g of water undergoes complete evaporation at 100ºC is_____.

(Given: ΔH_{vap} for water at 373K = 41kJ/ mol, R=8.314JK^{-1} mol^{-1})

Key: 189494

Sol:

$$H_2O_{(l)} \longrightarrow H_2O_{(g)}$$

$$\Delta H = \Delta u + \Delta n_g RT$$

$$= \Delta u + RT$$

$$\Delta u = \Delta H - RT$$

$$= 41000 - 8.314 \times 373$$

$$= 37898.878 \ \ Jmol^{-1}$$

For 5 mole $= \Delta u = 189494.39$

24. The Gibbs energy change (in J) for the given reaction at $\left[Cu^{2+}\right] = \left[Sn^{2+}\right] = 1M$ and 298 K is:

$Cu(s) + Sn^{2+}(aq) \longrightarrow Cu^{2+}(aq) + Sn(s);$

$\left(E^0_{Sn^{2+}|Sn} = -0.16V, E^0_{Cu^{2+}|Cu} = 0.34V,\ \text{Take F=96500 C mol}^{-1}\right)$

Key: 96500

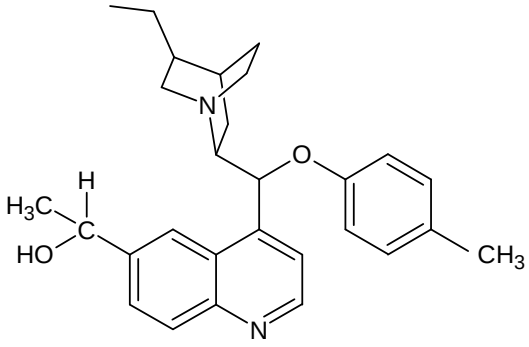
Sol:

$$E^\circ_{cell} = -0.16 - 0.34 = 0.50\,V$$

$$\Delta G^\circ_{cell} = -nFE^o_{cell}$$

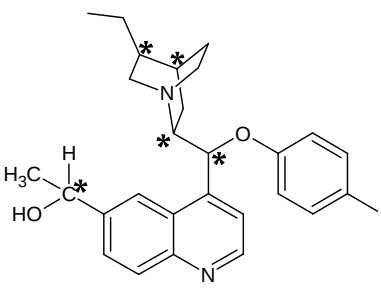
$$= -2 \times 96500 \times 0.5 = 96500 \ J$$

25. The number of chiral carbons present in the molecule given below is____



Key: 5

Sol:



MATHEMATICS

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)$ is

1) $\frac{1}{2}\left(1-i\sqrt{3}\right)$ 2) $\frac{1}{2}\left(\sqrt{3}-i\right)$

3) $-\frac{1}{2}\left(\sqrt{3}-i\right)$ 4) $-\frac{1}{2}\left(1-i\sqrt{3}\right)$

Key: 3

Sol: $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3 = \theta = \frac{\pi}{2}-\frac{2\pi}{9} = \frac{9\pi-415}{18}$

$= \left(\frac{1+\cos\theta+i\sin\theta}{1+\cos\theta-\sin\theta}\right)^3 = \left(\frac{\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}-i\sin\theta}\right)^3$ $= \frac{5\pi}{18}$

$= \left(\cos\theta+i\sin\theta\right)^3$ $3\theta = \frac{5\pi}{6}$

$= \cos 3\theta + i\sin 3\theta = \cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6} = \frac{-\sqrt{3}}{2}+i\frac{1}{2}$

$= -\frac{1}{2}\left(\sqrt{3}-i\right)$

2. Let $y = y(x)$ be the solution of the differential equation,

$\frac{2+\sin x}{y+1}.\frac{dy}{dx} = -\cos x, y > 0, y(0) = 1$. If $y(\pi) = a$ and $\frac{dy}{dx}$ at $x = \pi$ is b, then the ordered pair (a,b) is equal to

1) $(2,1)$ 2) $(1,-1)$ 3) $(1,1)$ 4) $\left(2,\frac{3}{2}\right)$

Key: 3

Sol: $\int \frac{dy}{y+1} = -\int \frac{\cos x}{\sin x + 2} dx$

$\Rightarrow \ln|y+1| = -\ln|\sin x + 2| + c$

$\Rightarrow \ln 2 = -\ln 2 + c \quad (y(0) = 1)$

$\Rightarrow C = 2\ln 2$

$\therefore \ln|y+1| = -\ln|\sin x + 2| + 2\ln 2$

$\Rightarrow \ln(y+1) = -\ln(\sin x + 2) + 2\ln 2$

$\Rightarrow y + 1 = \frac{4}{\sin x + 2}$ $\qquad y(B) = a \Rightarrow a + 1 = 2 \qquad a \Rightarrow 1$

$\therefore (a, b) = (1,1) \qquad \frac{dy}{dx} = \frac{-4}{(\sin x 2)^2}.\cos x$

$x = \pi \Rightarrow b = \frac{4}{4} = 1$

b = 1

3. If a function $f(x)$ defined by $f(x) = \begin{cases} ae^2 + be^{-1} & , \quad -1 \le x < 1 \\ cx^2 & , \quad 1 \le x \le 3 \\ ax^2 + 2cx & , \quad 3 < x \le 4 \end{cases}$ be continuous for some $a, b, c \in R$ and $f'(0) + f'(2) = e$, then the value of a is

1) $\frac{1}{e^2 - 3e + 13}$ 2) $\frac{e}{e^2 + 3e - 13}$

3) $\frac{e}{e^2 - 3e + 13}$ 4) $\frac{e}{e^2 + 3e + 13}$

Key: 3

Sol: $ae + \frac{b}{e} = c$ $\qquad 9c = 9a + bc$ $\qquad f'(0) + f'(2) = e$

$\Rightarrow 3c = 9a$ $\qquad \Rightarrow a - b + 4c = e$

$\Rightarrow c = 3a$ $\qquad \Rightarrow a - b + 12a = e$

$\Rightarrow 13a - b = e$

Now (1) $\Rightarrow ae + \frac{1}{e}(13a - e) = 3a$

$\Rightarrow a\left(e + \frac{13}{e} - 3\right) = 1$ $\qquad \Rightarrow a = \frac{e}{e^2 - 3e + 13}$

4. If $P(x)$ be a polynomial of degree three that has a local maximum value 8 at x=1 and a local minimum value 4 at x=2; then p(0) is equal to

1) 12 2) -12 **3)** -24 4) 6

Key: 2

Sol: $p^1(x) = a(x-1)(x-2)$

$\therefore p(x) = a\left(\frac{x^3}{3} - \frac{3x^2}{2} + 2x\right) + C \quad \begin{matrix} p(1)=8 \\ p(2)=1 \end{matrix} \quad \begin{matrix} \frac{5}{6}a+c=8 \\ \frac{2}{3}a+c=4 \end{matrix}$

$p(0) = e \qquad \Rightarrow p(0) = -12 \qquad -\frac{5}{4}(2) \Rightarrow c\left(1-\frac{5}{4}\right) = 3$

C =-12

5. If $R = \{(x,y) : x,y \in Z, x^2 + 3y^2 \le 8\}$ is a relation on the set of integers Z, then the domain of R^{-1} is:

1) $\{-2,-1,1,2\}$ 2) $\{-1,0,1\}$ 3) $\{0,1\}$ 4) $\{-2,-1,0,1,2\}$

Key: 2

Sol: $R = \{(0,0),(1,0),(2,0),(-1,0)(-2,0)$

$\begin{Bmatrix} (1,1) & (2,1) \\ (-1,1) & (-2,1) \\ (-1,-1) & (-3,-1) \\ (1,-1) & (2,-1) \end{Bmatrix}$

Range of R = Done of $R^{-1} = \{0,1,-1\}$

6. The sum of the first three terms of a GP. Is S and their product is 27. Then all such S lie in.

1) $(-\infty, 9]$ 2) $(-\infty,-3] \cup [9,\infty)$

3) $(-\infty,-9] \cup [3,\infty)$ 4) $[-3,\infty)$

Key: 2

Sol: $\frac{a}{r}, a, ar$ $\quad a\left(\frac{1}{r}+1+r\right)=S$ $\quad$ product $=a^3=27\Rightarrow a=3$

$\therefore 3\left(r+\frac{1}{r}+1\right)=S$

$\therefore S\in(-\infty,-3]\cup[9,\infty)$

7. If the tangent to the curve $y=x+\sin y$ at a point (a,b) is parallel to the line joining $\left(0,\frac{3}{2}\right)$ and $\left(\frac{1}{2},2\right)$, then :

1) $b=a$ 2) $|b-a|=1$ 3) $b=\frac{\pi}{2}+a$ 4) $|a+b|=1$

Key: 2

Sol: $\frac{dy}{dx}=1+\cos y.\left(\frac{dy}{dx}\right)$ $\quad\therefore$ slope = slope

$\Rightarrow \frac{dy}{dx}=\frac{1}{1-\cos y} \quad \Rightarrow \frac{1}{1-\cos b}=\frac{2-3}{\frac{1}{2}-0}$

$\text{slope}=\frac{1}{1-\cos b} \quad \Rightarrow 1-\cos b=1\Rightarrow \cos b=0$

$\therefore b=2\alpha\pi\pm\frac{\pi}{2}$

Also $b=a+\sin b$

$\Rightarrow b-a=\sin b$

$\Rightarrow b-a=\pm 1$

$\Rightarrow |b-a|=1$

8. Let $\alpha>0,\beta>0$ be such that $\alpha^3+\beta^2=4$. If the maximum value of the term independent of x in the binomial expansion of $\left(\alpha x^{\frac{1}{9}}+\beta x^{-\frac{1}{6}}\right)$ is 10k, then k is equal to

1) 84 2) 176 3) 352 4) 336

Key: 4

Sol: Let . If the maximum value of the term side pendat of x in the binomial expansion of

$\left(\alpha x^{1/9}+\beta x^{-1/6}\right)^{10-r}.\left(\beta x^{-1/6}\right)^{r} \Rightarrow \frac{10-r}{9}-\frac{r}{6}=0$

$\Rightarrow 20-2r-3r=0$

$\Rightarrow r=4$

$\therefore 10K = {}^{10}C_4.\alpha^6.\beta^4$

$\Rightarrow 10K \le \frac{10\times 9\times 8\times 7}{24}\times 16$

$\Rightarrow k \le 21\times 16$

$\Rightarrow k \le 336$

$K = 336$

$\frac{\alpha^2+\beta^2}{2} \ge \sqrt{\alpha^3\beta^2}$

$\Rightarrow \frac{4}{2} \ge \sqrt{\alpha^3\beta^2}$

$\Rightarrow \alpha^6\beta^4 \le 16$

9. Let A be a 2×2 real matrix with entries from {0,1} and $|A| \neq 0$. Consider the following two statements:

P) If $|A| \neq I_2$ then $|A| \neq -1$

Q) If $|A| = 1$ then $\text{tr}(A) = 2$

Where I_2 denotes 2×2 identity matrix and tr(A) denotes the sum of the diagonal entries of A. Then.

1) P is false and Q is true
2) Both P and Q are true
3) Both P and Q are false
4) P is true and Q is false

Key: 1

Sol: $P: \text{Let } A=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} \Rightarrow |A|=1 \quad \therefore P \text{ is false}$

$Q: \text{Let } |A|=1 \Rightarrow A\begin{pmatrix}1 & x\\ y & 1\end{pmatrix} \Rightarrow \text{trace}A = 1 \Rightarrow Q \text{ true}$

10. If $|x|<1, |y|<1$ and $x \neq y$, then the sum to infinity of the following series $(x+y)+\left(x^2+xy+y^2\right)+\left(x^3+x^2y+xy^2+y^3\right)+....$ is

1) $\frac{x+y-xy}{(1-x)(1-y)}$
2) $\frac{x+y+xy}{(1+x)(1+y)}$
3) $\frac{x+y+xy}{(1-x)(1-y)}$
4) $\frac{x+y-xy}{(1-x)(1+y)}$

Key: 1

Sol: $GE=(x+y)+(x^2+xy+y^2)+(x^3+x^2y+xy^2+y^3)t.....$

$$=\frac{1}{(x-y)}\left\{(x^2-y^2)+(x^3-y^3)+(x^4-y^4)+.....\right\}$$

$$=\frac{1}{(x-y)}\left\{\frac{x^2}{1-x}-\frac{y^2}{1-y}\right\} \qquad (\because \text{Infinity GPs})$$

$$=\frac{1}{x-y}\left\{\frac{x^2-x^2y-y^2+x^2y}{(1-x)(1-y)}\right\}=\frac{1}{x-y}\frac{(x+y)(x-y)+xy(x-y)}{(1-x)(1-y)}$$

$$=\frac{x+y+xy}{(1-x)(1-y)}$$

11. Box I contains 30 cards numbered 1 to 30 and box II contains 20 cards numbered 31 to 50. A box is selected at random and a card is drawn from it. The number on the card is found to be a non – prime number. The probability that the card was drawn from Box I is

1) $\frac{2}{3}$ 2) $\frac{2}{5}$ 3) $\frac{4}{17}$ 4) $\frac{8}{17}$

Key: 4

Sol: 1,2,3, …….30 I 31, 32, ……., 50 II

$$R.P=\frac{\frac{1}{2}\times\frac{20}{30}}{\frac{1}{2}\times\frac{20}{30}+\frac{1}{2}\times\frac{15}{20}}=\frac{\frac{2}{3}}{\frac{2}{3}+\frac{3}{4}}=\frac{2}{3}\times\frac{12}{14}=\frac{8}{17}$$

12. The domain of the function $f(x)=\sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$ is $(-\infty,-a]\cup[a,\infty)$. Then a is equal to

1) $\frac{1+\sqrt{17}}{2}$ 2) $\frac{\sqrt{17}-1}{2}$ 3) $\frac{\sqrt{17}}{2}+1$ 4) $\frac{\sqrt{17}}{2}$

Key: 1

Sol: $f(x)=\sin^{-1}\left(\frac{|x|-5}{x^2+1}\right)\Rightarrow -1\le\frac{|x|<5<1}{x^2+1} \quad |x|=1$

$\Rightarrow t+5\le t^2+1 \qquad \Rightarrow t^2-t-4\ge 0$

$\frac{1\pm\sqrt{17}}{2}\Rightarrow|x|\ge\frac{1+\sqrt{17}}{2}\Rightarrow|x|\ge a\Rightarrow a=\frac{1+\sqrt{17}}{2}$

$-t^2-t\le t+5\Rightarrow t^2+t+6\ge 0\Rightarrow +\in R\Rightarrow|x|\in R\Rightarrow x\in R$

13. A line parallel to the straight line $2x-y=0$ **is tangent to the hyperbola** $\frac{x^2}{4}-\frac{y^2}{2}=1$ at the point (x_1,y_1). Then $x_1^2+5y_1^2$ is equal

1) 5 **2)** 10 3) 8 4) 6

Key: 4

Sol: $\frac{x^2}{4}-\frac{y^2}{2}=1$, tgt ll to $2x-y=0\Rightarrow 2x-y+c=0$

Tgt at $\left(x_1y_1\Rightarrow S_1=0\Rightarrow \frac{xx_1}{4}-\frac{yy_1}{2}-1=0\right)$ $C^2=a^2x^2-b^2$

$=4(2)^2-2$

$=14$

$C=\pm\sqrt{14}$

$\frac{x_1}{\frac{4}{2}}=\frac{-\frac{y_1}{2}}{-1}=\frac{-1}{c}$

$\Rightarrow \frac{x_1}{8}=\frac{y_1}{2}=\frac{-1}{\pm\sqrt{14}}$

$\Rightarrow x_1=\pm 8\sqrt{14}, y_1=\pm 2/\sqrt{14}$

$\Rightarrow GE=\left(8/\sqrt{14}\right)^2+5\left(2\sqrt{14}\right)^2=\frac{64}{14}+5\left(\frac{4}{14}\right)=\frac{84}{14}=6$

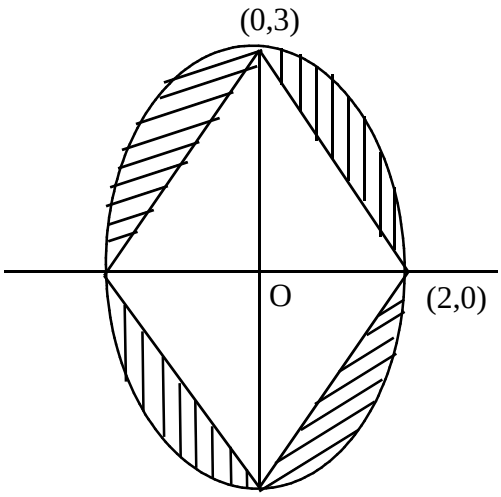
14. Area (in sq. units) of the region outside $\frac{|x|}{2}+\frac{|y|}{3}=1$ and inside the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ **is**

1) $3(4-\pi)$ 2) $6(4-\pi)$ 3) $6(\pi-2)$ 4) $3(\pi-2)$

Key: 3

Sol: $\frac{|x|}{2}+\frac{|y|}{3}=1, \frac{x^2}{4}+\frac{y^2}{9}=1$

Repcrea = Ellipse – rhombus



$= \pi(2)(3) - 4\left(\frac{1}{2} \times 2 \times 3\right)$

$= 6\pi - 12$

$= 6(\pi - 2)$

15. Let $X = \{x \in N : 1 \le x \le 17\}$ and $Y\{ax + b; x \in X \text{ and } a, b \in R, a > 0\}$. If mean and variance of elements of Y are 17 and 216 respectively then a+b is equal to :

1) 7 2) -7 3) 9 4) -27

Key: 2

Sol: Let $X = \{x \in N / 1 \le x \le 17\}$

$Y = \{ax + b / x \in N, a, b \in R, a > 0\}$

For X : Mean of $= \frac{1 + 2 + ... + 17}{17} = \frac{17 \times 18}{2 \times 17} = 9$

Mean of $\Rightarrow 9a + b = 17$ _____ (1)

Varie of X $= \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x_i}{n}\right) = \frac{17 \times 18 \times 35}{6 \times 17} - (9)^2$

$= 105 - 81 = 24$

Varie of Y $= a^2(24) = 216 \Rightarrow a^2 = 9 \Rightarrow a = 3$

$\therefore b = -10$

$a + b = -7$

16. The contrapositive of the statement If I reach the station in time, then I will catch the train is

1) If I will not catch the train, then I do not reach the station in time.

2) If I do not reach the station in time, then I will catch the train.

3) If I will catch the train, then I reach the station in time

4) If I do not reach the station in time, then I will not catch the train

Key: 1

Sol: p = I nado station in time
q = I will catch the train
We have to find condra positive of "$p \to q$"
i.e $\sim q \to \sim p$
ie If I will not catch the train then I donot reach the station in time.

17. The Plane passing through the points (1,2,1) , (2,1,2) and parallel to the line, $2x = 3y, z = 1$ also passes through the point:

1) $(0,-6,2)$ 2) $(-2,0,1)$ 3) $(0,6,-2)$ 4) $(2,0,-1)$

Key: 2

Sol: Plane through $(1,2,1),(2,1,2)$ parallel to line $2x = 3y, z = 1$

$$\begin{matrix} 2 & -3 & 0 \\ 0 & 0 & 1 \end{matrix}$$

-3 -2 0 (disc of line) $A(1,2,1)\,B(2,1,2)$

$\therefore$ drS of $AB: 1,-1,1$

$$\therefore \begin{matrix} -3 & -2 & 0 \\ 1 & -1 & 1 \end{matrix}$$

-2 3 5 $\therefore 2x - 3y - 5z = 2 - 6 - 5$

$\Rightarrow 2x - 3y - 5z = -9$

Passes to (-2, 0, 1)

18. Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n, n = 1,2,3,....$ then.

1) $5S_6 + 6S_5 = 2S_4$ 2) $5S_6 + 6S_5 = 2S_4 = 0$

3) $5S_6 + 5S_5 = 2S_4 = 0$ 4) $6S_6 + 5S_5 = 2S_4$

Key: 1

Sol: $5x^2+6x-2=0<\begin{matrix}\alpha\\ \beta\end{matrix}$

Where, $ax^2+bx+c=0$

$\Rightarrow a(S_n)+b(S_{n-1})+C.S_{n-2}=0$

$5.S_6+6.S_5-2.S_4=0$

19. Let S be the set of all $\lambda \in R$ for which the system of linear equations

$2x-y+2z=2$

$x-2y+\lambda z=-4$

$x+\lambda y+z=4$

Has no solution. Then the set S

1) contains exactly two elements

2) contains more than two elements.

3) is singleton

4) is an empty set

Key: 1

Sol: $2x-y+2z=2$

$x-2y+\lambda z=-4$

$x+\lambda y+z=4$ has no solution.

$$\begin{vmatrix}2 & -1 & 2\\ 1 & -2 & \lambda\\ 1 & \lambda & 1\end{vmatrix}=0\Rightarrow\begin{vmatrix}0 & -1 & 2\\ 1-\lambda & -2 & \lambda\\ 0 & \lambda & 1\end{vmatrix}=0$$

$\Rightarrow -(1-\lambda)(-1-2\lambda)=0\Rightarrow \lambda=1,\lambda=-\frac{1}{2}$

For these values, the system is inconsistent

$\therefore$ No.of values of λ is 2

20. Let $P(h,k)$ be a point on the curve $y=x^2+7x+2$, nearest to the line, $y=3x-3$. Then the equation of the normal to the curve at P is

1) $x+3y+26=0$ 2) $x-3y-11=0$

3) $x-3y+55=0$ 4) $x+3y-62=0$

Key: 1

Sol: $y = x^2 + 7x + 2,\ y = 3x - 3$

$$\frac{dy}{dx} = 2x + 7 = 3 \Rightarrow x = -2$$

$$y = -8$$

$$(-2, -8)$$

$$N.S = \frac{-1}{2(-2)+7} = -\frac{1}{3} \Rightarrow \text{Eqn} : x + 3y = -26$$

(NUMERICAL VALUE TYPE)

This section contains 5 questions. Each question is numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.(e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $|\vec{a} - \vec{b}|^2 + |\vec{a} - \vec{c}|^2 = 8$. Then $|\vec{a} + 2\vec{b}|^2 + |\vec{a} + 2\vec{c}|^2$ is equal to _____

Key: 2.00

Sol: $|\bar{a}| = 1, |5| = 1, |\bar{c}| = 1$

$$|\bar{a} - \bar{b}|^2 + |\bar{a} - \bar{c}|^2 = 8$$

$$\Rightarrow 1 + 1 - 2\bar{a}.\bar{b} + 1 + 1 - 2\bar{a}.\bar{c} = 8$$

$$\Rightarrow \bar{a}.\bar{b} + \bar{a}.\bar{c} = -2$$

$$\therefore GE = 2$$

$$|\bar{a} + 2\bar{b}|^2 + |\bar{a} + 2\bar{c}|^2$$

$$= 1 + 4 + 4\bar{a}.\bar{b} + 1 + 4 + 4\bar{a}.\bar{c}$$

$$= 10 + 4(-2)$$

$$= 2$$

22. The integral $\int_0^2 \big||x-1| - x\big| dx$ is equal to

Key: 1.50

Sol: $\int_0^2 \big||x-1| - x\big| dx$

$$= \int_0^1 |1 - 2x| dx + \int_1^2 |(-1)| dx$$

$$= \int_0^1 (1 - 2x) dx + \int_{1/2}^1 (2x - 1) dx + 1$$

$$= \left(x - x^2\right)_0^{1/2} + \left(x^2 - x\right)_{1/2}^1 + 1 = 1.5$$

23. If the letters of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then the position of the word 'M' is_____

Key: 309.00

Sol: MOTHER EHMORT

$\text{Rank} = (2)5! + (2)4! + (3)3! + (1)2! + (0)1! + 1$

$= 240 + 48 + 18 + 2 + 1 = 309$

24. The number of integral values of k for which the line, $3x + 4y = k$ intersects the circle, $x^2 + y^2 - 2x - 4y + 4 = 0$ at two distinct points is_____

Key: 9.00

Sol: $3x + 4y = k$ $x^2 + y^2 - 2x - 4y + 4 = 0$

$e = (1,2)$

$\text{rad} = 1$

$d < r$

$\Rightarrow \frac{|1 - K|}{5} < 1 \Rightarrow |11 - K| < 5$

$\Rightarrow -5 < K - 11 < 5$

$\Rightarrow 6 < K < 16$

$\Rightarrow K = 7,8,9,.....,15$

No of integers = 9

25. If $\lim\limits_{x \to 1} \frac{x + x^2 + x^3 + ... + x^n - n}{x - 1} = 820, (n \in N)$ then the value of n is equal to _____

Key: 40.00

Sol: $\lim\limits_{x \to 1} \frac{x + x^2 + x^n - n}{x - 1} \left(\frac{0}{0}\right)$

LH role $\Rightarrow \lim\limits_{x \to 1} \frac{1 + 2x + 3x^2 + h.x^{n-1} - 0}{1}$

$= 1 + 2 + 3 + ... + n = \frac{n(n+1)}{2} = 820(\text{given}) \Rightarrow n = 40$

Prepared by

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